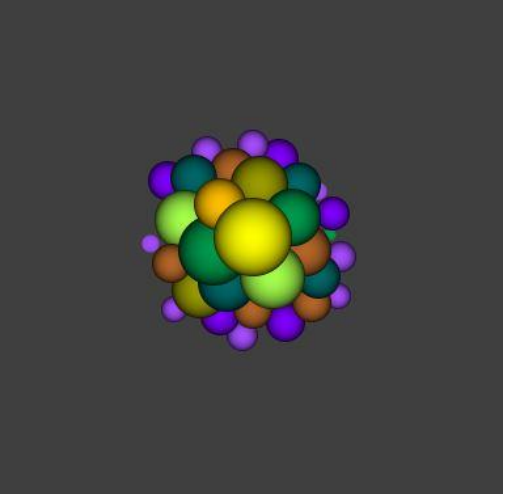
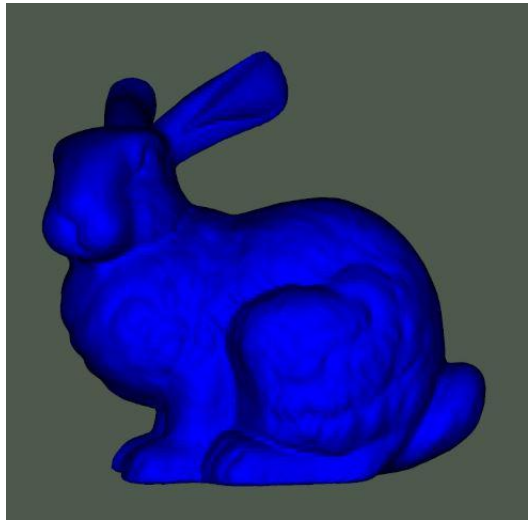


CS 440: Computer Graphics

Week 6

Lecture 11: Mesh Data Structures



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

- Quiz next week on Monday
- Extra Office Hours: Wed 3:30pm-5:00pm

- Recap:
  - Concatenating Transformations
  - General Transformations
  - Model Instance Paradigm
  - Scene Graphs
  - 3D Geometry
  - Topology

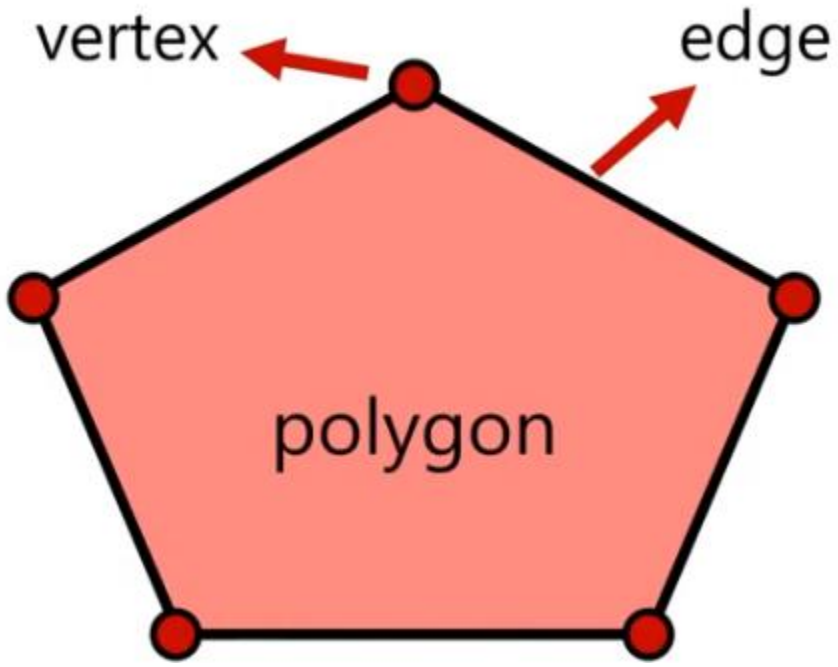
Today: Mesh Data Structures

- Topology
- Mesh Data Structures
  - Face Set
  - Shared Vertex
  - Adjacency Matrix
  - Winged Edge
  - Half-edge

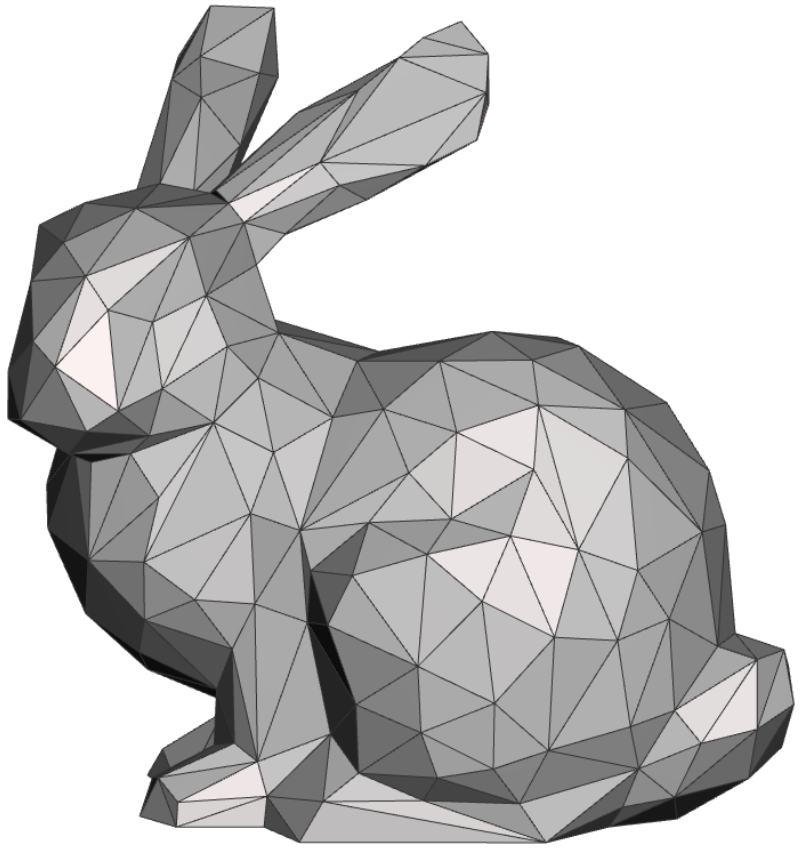
Recall: 3D Meshes

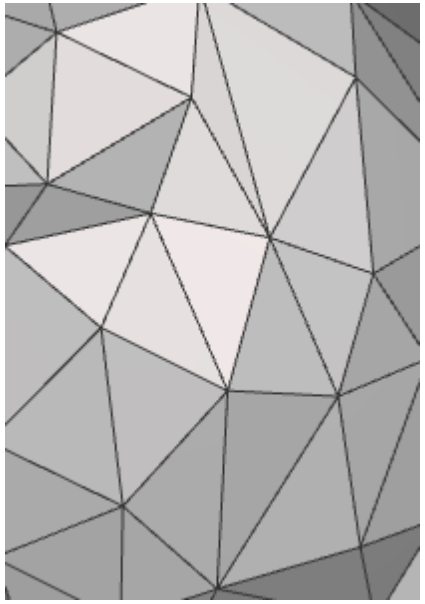
- What are 3D meshes made of?
  - Vertices
  - Edges
  - Faces (or polygons, usually triangles)

- Meshes represent surfaces



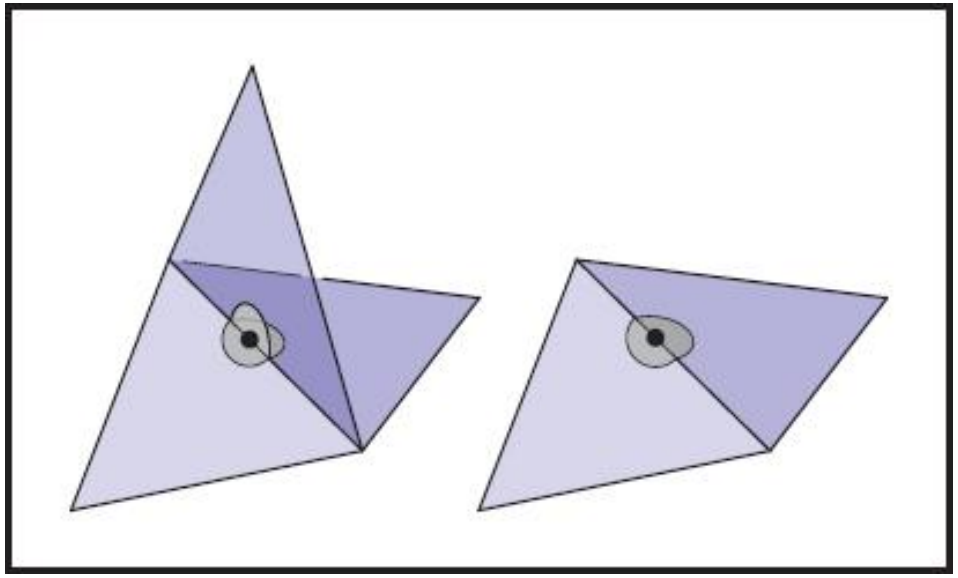
3D Meshes

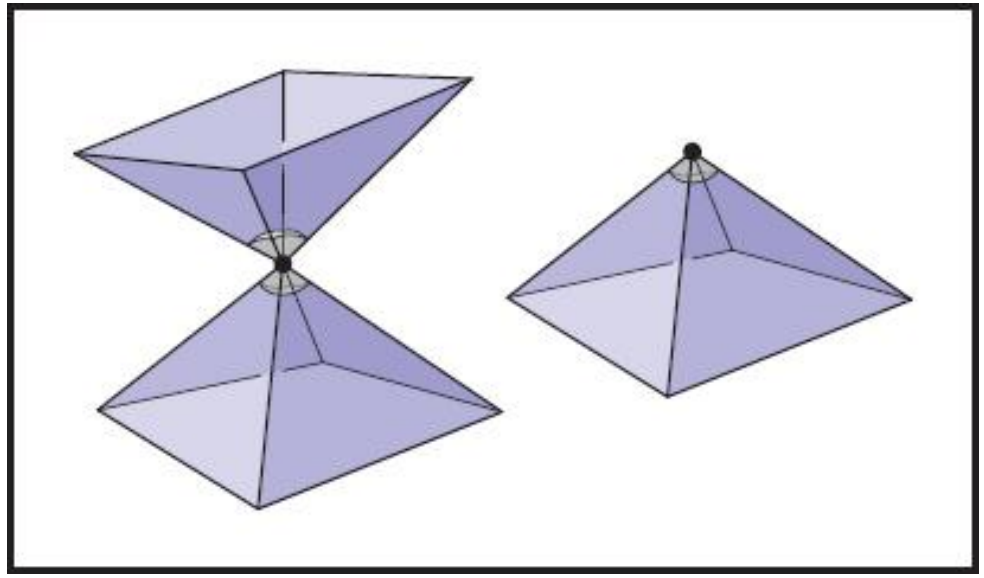




Mesh Topology: Manifold vs Non-manifold

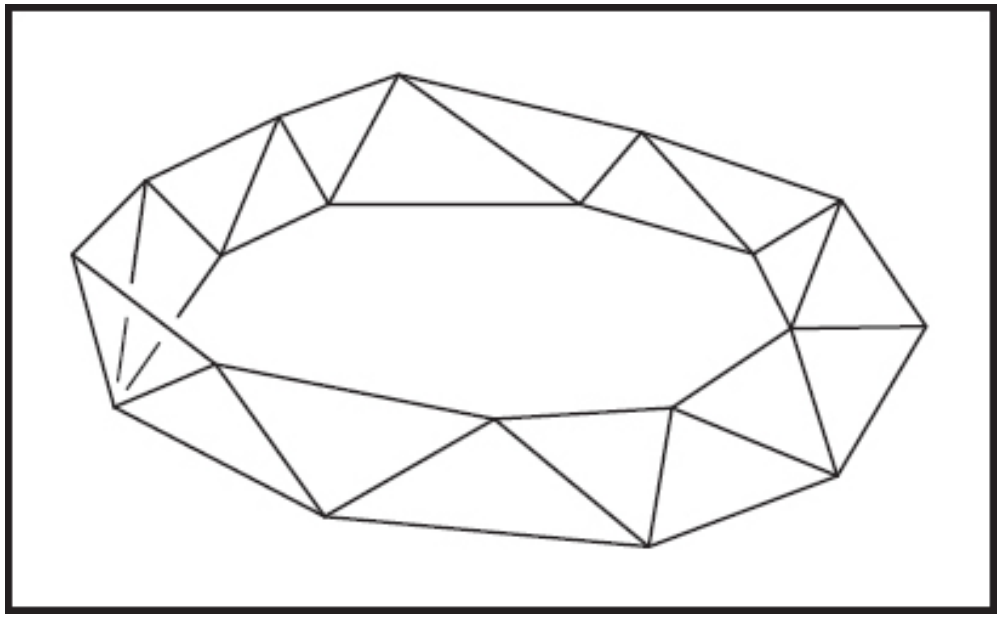
- Edge manifold
- Vertex Manifold





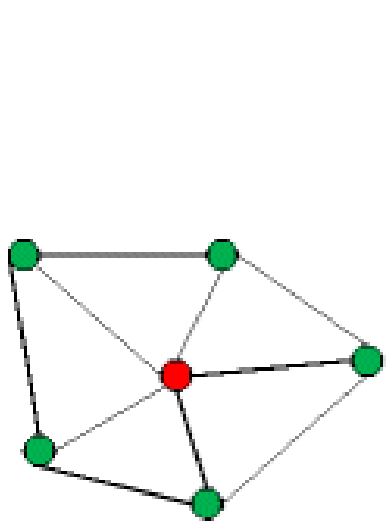
Mesh Topology: Boundary Edges

- Meshes can have **boundaries**
- The can still be **manifold meshes with boundary**

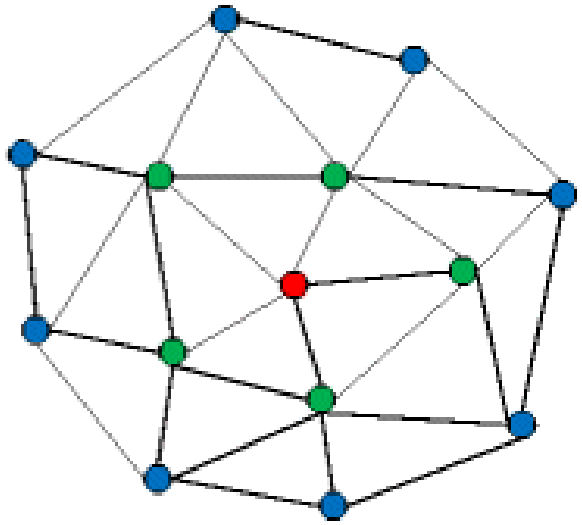


One-ring neighbors

- Local neighborhood of a vertex
- All the vertices that share a common face with a specific vertex are its **one-ring neighbors**



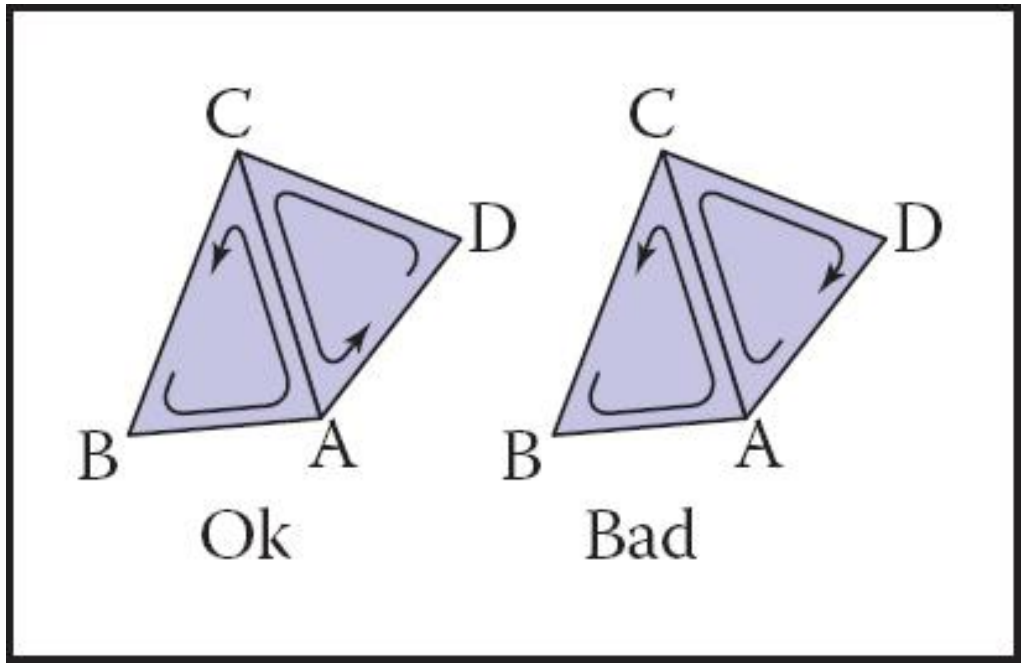
(a) 1-ring



(b) 2-ring

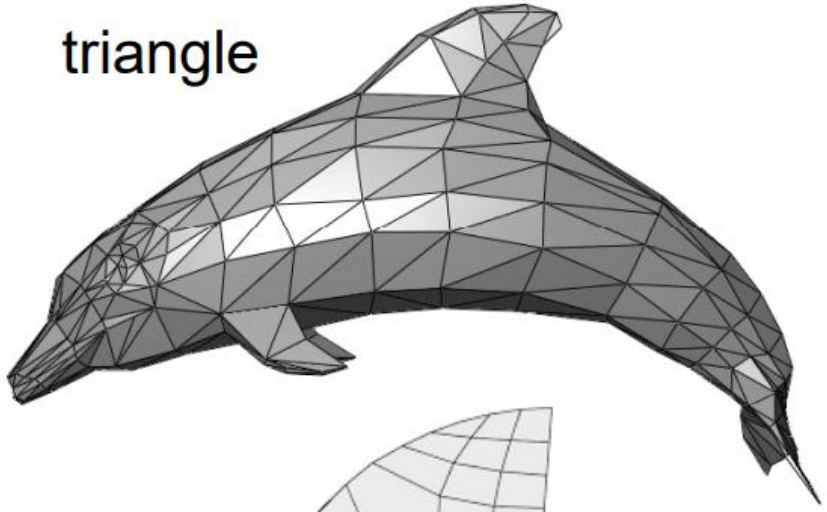
Mesh Topology: Orientation

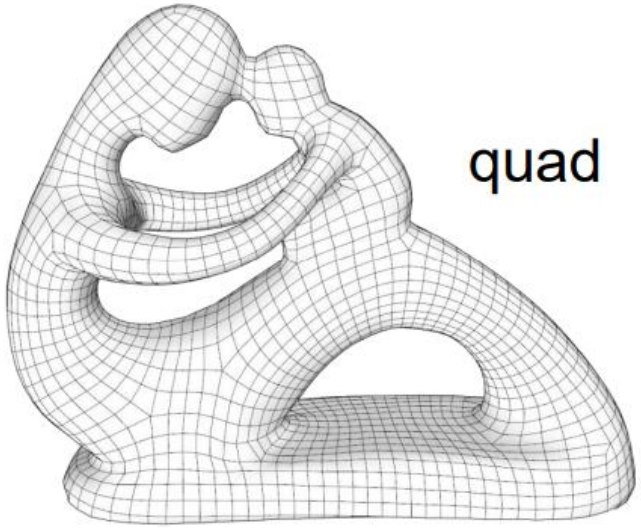
- We need to distinguish between the “front” and “back” of a face
- Orientation depends on the order of listed vertices
- A mesh is **consistently oriented** if all triangles are defined in the same order

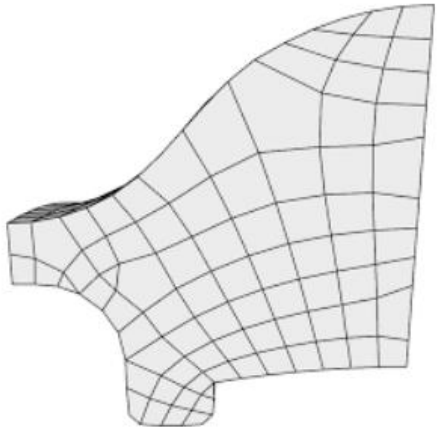


Obtained from https://www.cs.utexas.edu/~fussell/courses/cs384g/lectures/lecture19.pdf

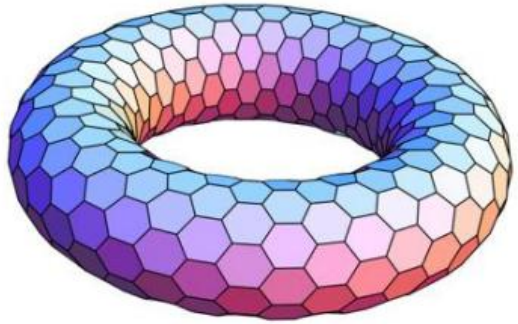
Polygonal Meshes







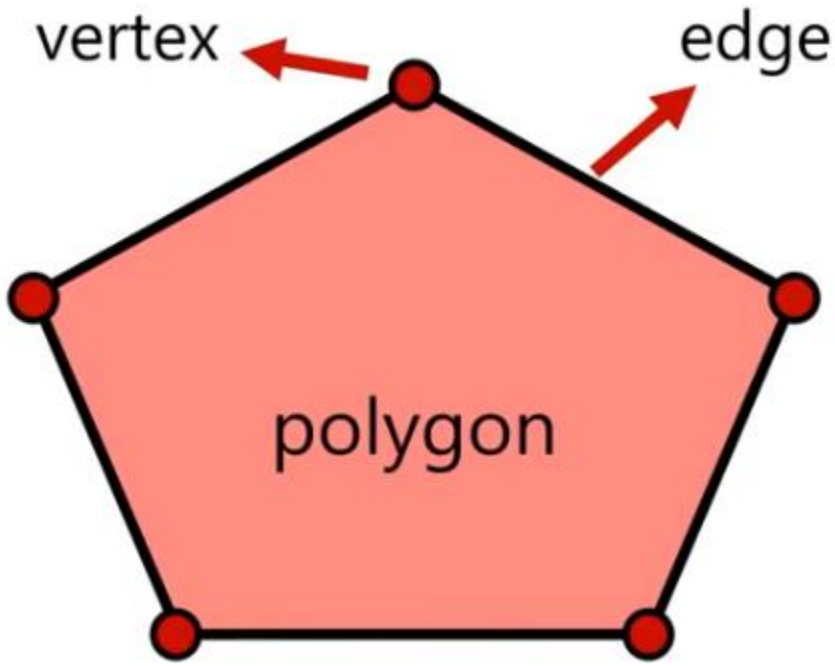
quad-dominant



exotic (hexagonal, etc)

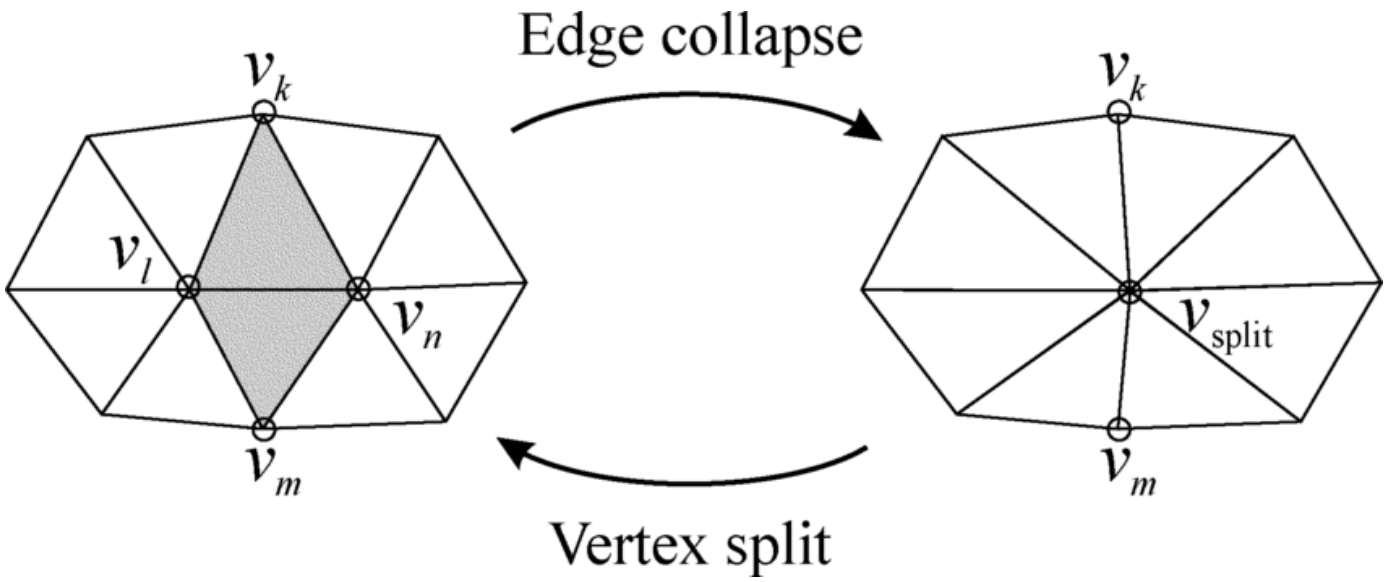
3D Meshes: What to Store?

- What information do we store for meshes?
  - Geometry: Vertices
  - Connectivity: Edges
  - Attributes: Colors, normals, …



Mesh Connectivity

- What kind of operations or queries should be supported?
  - Adjacency information per vertex/edge/face
  - Modification: Addition and Removal of vertex/edge/face



Mesh Connectivity

- **What kind of operations or queries should be supported?**
- Face → Vertices
- Face → Face
- Vertex → Face
- Vertex → Vertex

Mesh Data Structure Considerations

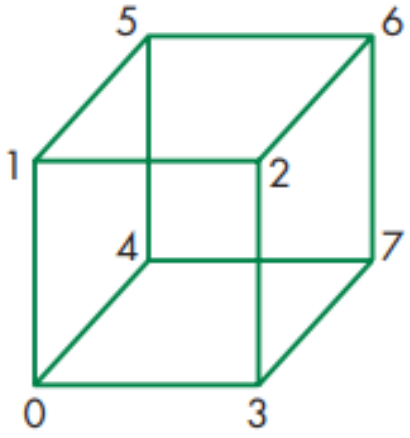
- Time to build
- Time to process a query
- Time to perform an operation
- Space complexity
- Redundancy

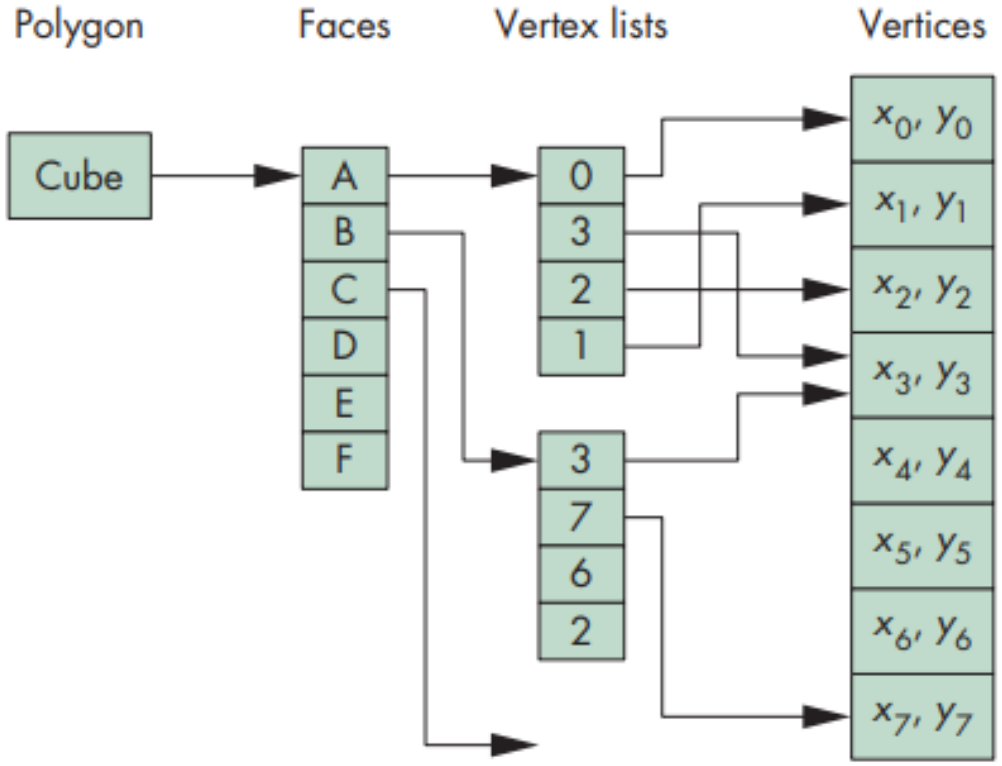
Face Set

- Also known as “polygon soup”
- List of faces – simple!
- Format used in .CLI and .STL files

```
begin
vertex x y z
vertex x y z
end
```

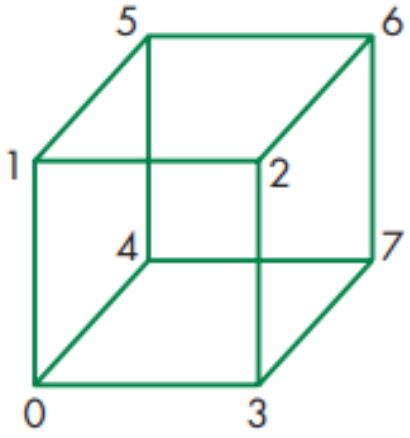
Face Set

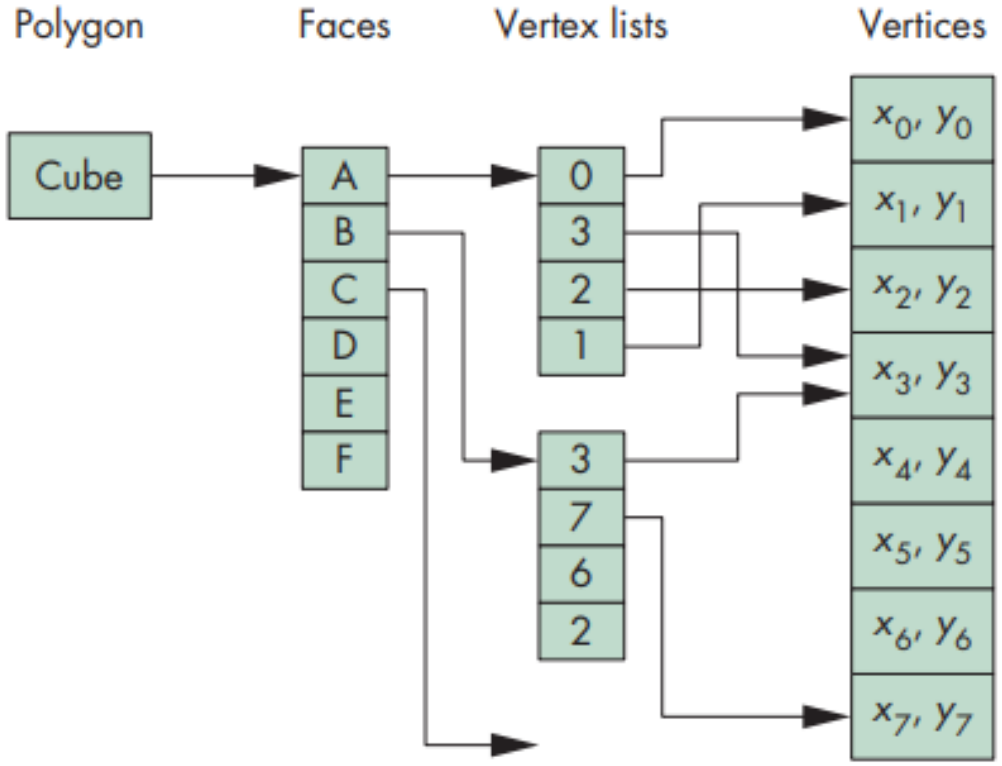

5
6
1
2
4
7
0
3


Polygon
Faces
Vertex lists
Vertices
Cube
A
B
C
D
E
F
0
3
2
1
3
7
6
2
x_0, y_0
x_1, y_1
x_2, y_2
x_3, y_3
x_4, y_4
x_5, y_5
x_6, y_6
x_7, y_7

Face Set

Drawbacks?





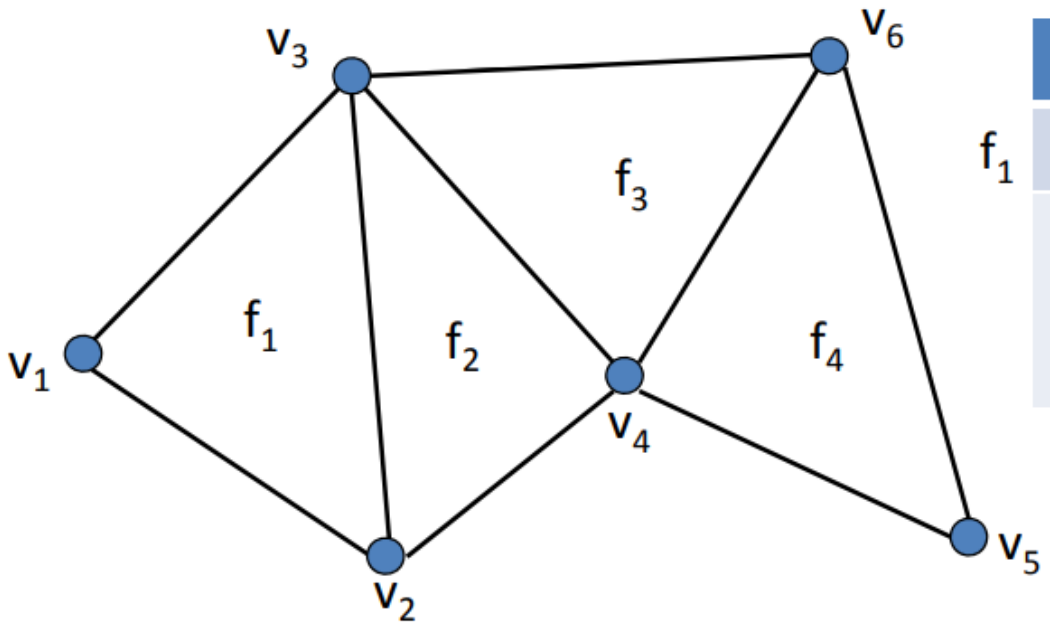
Shared Vertex

- "Face-vertex lists" or "Indexed Meshes"
- List of faces **that refer to vertices**
- Format used in .OBJ and .PLY files

```
v -1 -1 -1   → 0
v -1 1 1     → 1
v 1 -1 1     → 2
v 1 1 1      → 3
...
f 0 1 2
f 0 2 3
f 0 3 1
f 2 1 3
...
```

Implied vertex indices

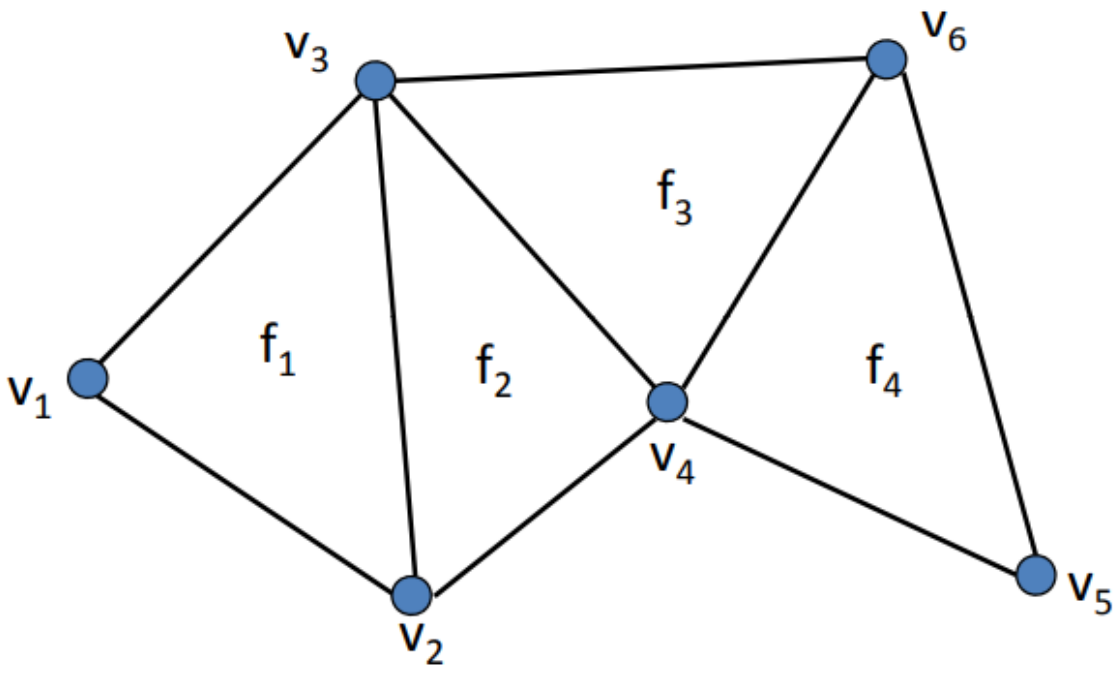
Shared Vertex



	TRIANGLES		
f_1	2	3	1
		.	
		.	
		.	

	VERTICES
v_1	[20 10 0]
v_2	[19 20 0]
v_3	[14 15 0]
	.
	.
	.

Adjacency Matrix

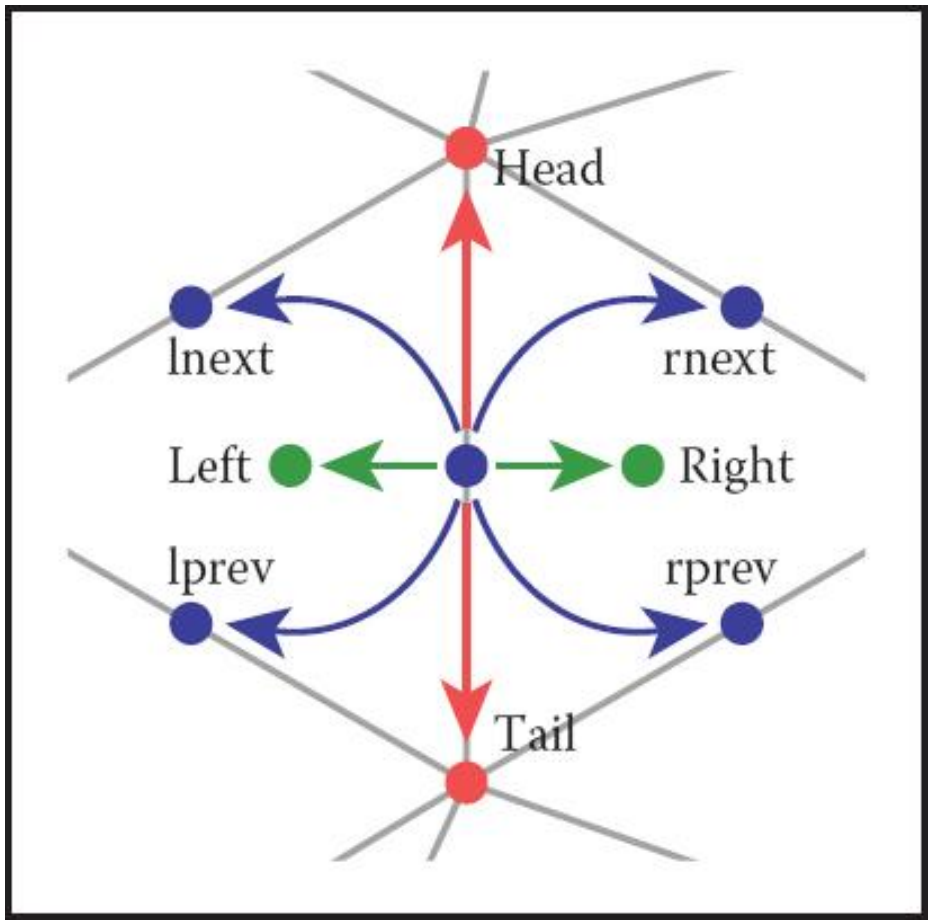


A =

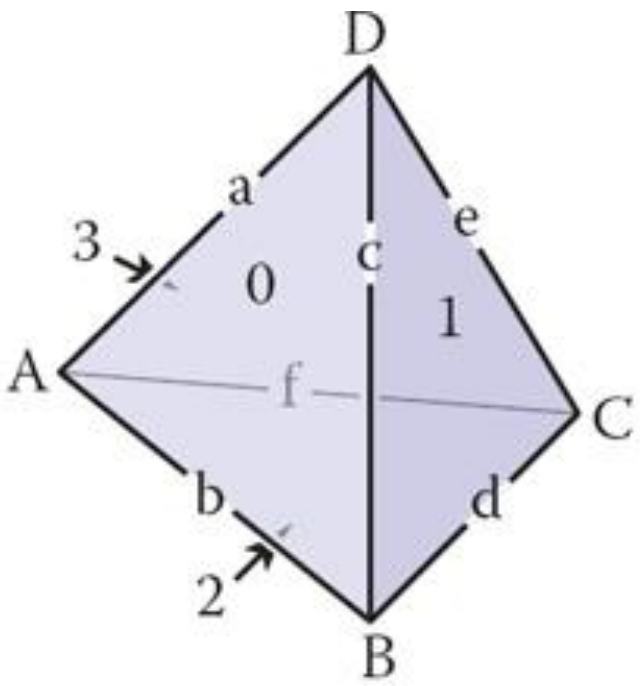
	v_1	v_2	v_3	v_4	v_5	v_6
v_1		1	1			
v_2	1		1	1		
v_3	1	1		1		1
v_4		1	1		1	1
v_5				1		1
v_6			1	1	1	

Winged Edge

- Each edge *e* stores pointers to:
  - The **head** and **tail** vertices (vertices that make the edge *e*)
  - **Left** and **right** faces (the faces that share the edge *e*)
  - The **next** and **previous** edges



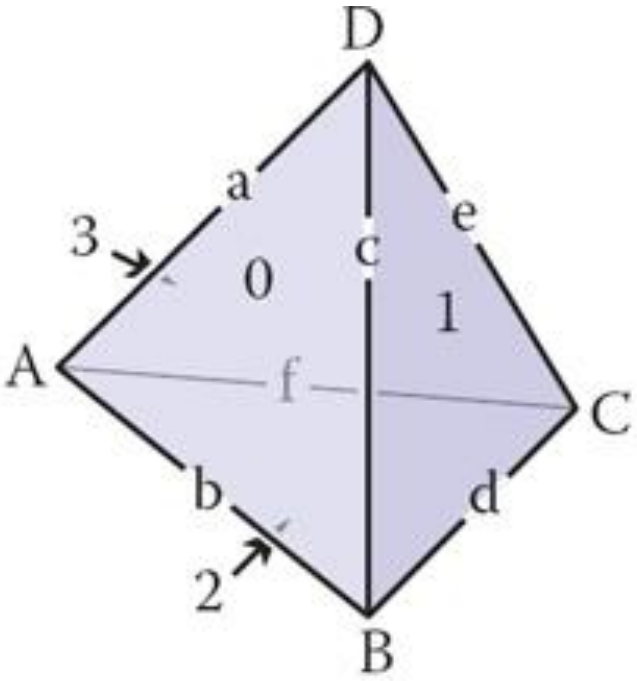
Winged Edge



Edge	*Vertex 1*	*Vertex 2*	*Face left*	*Face right*	*Pred left*	*Succ left*	*Pred right*	*Succ right*
a	A	D	3	0	f	e	c	b
b	A	B	0	2	a	c	d	f
c	B	D	0	1	b	a	e	d
d	B	C	1	2	c	e	f	b
e	C	D	1	3	d	c	a	f
f	C	A	3	2	e	e	b	d

Winged Edge

- Each **vertex** stores a pointer to one of its adjacent edges
- Each **face** stores a pointer to one (and any) of its edges

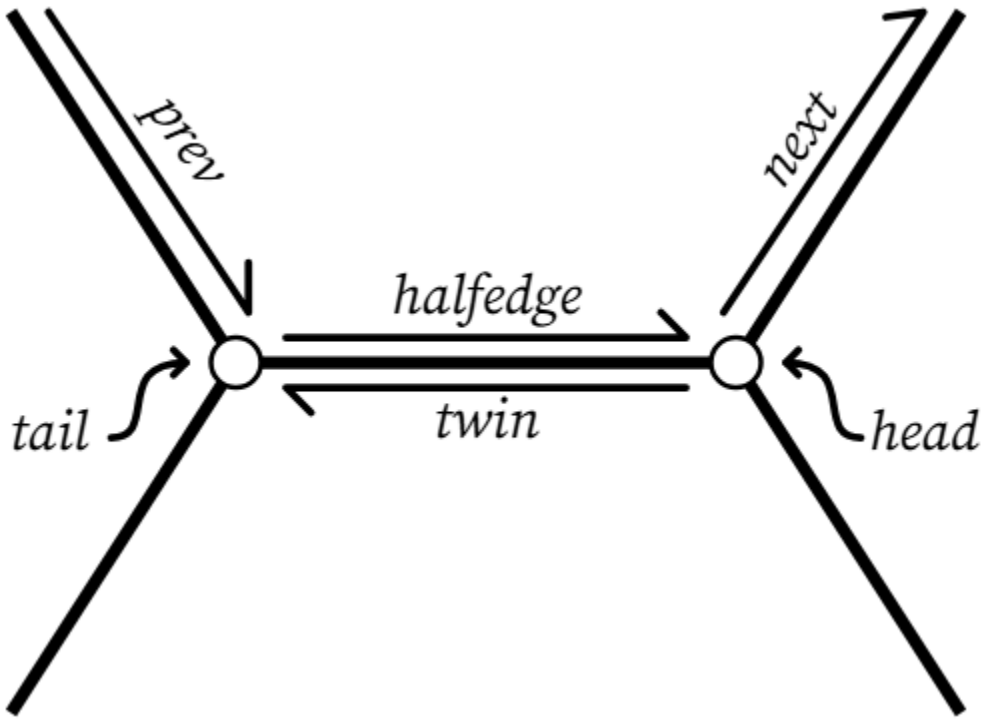


Vertex	*Edge*
A	a
B	d
C	d
D	e

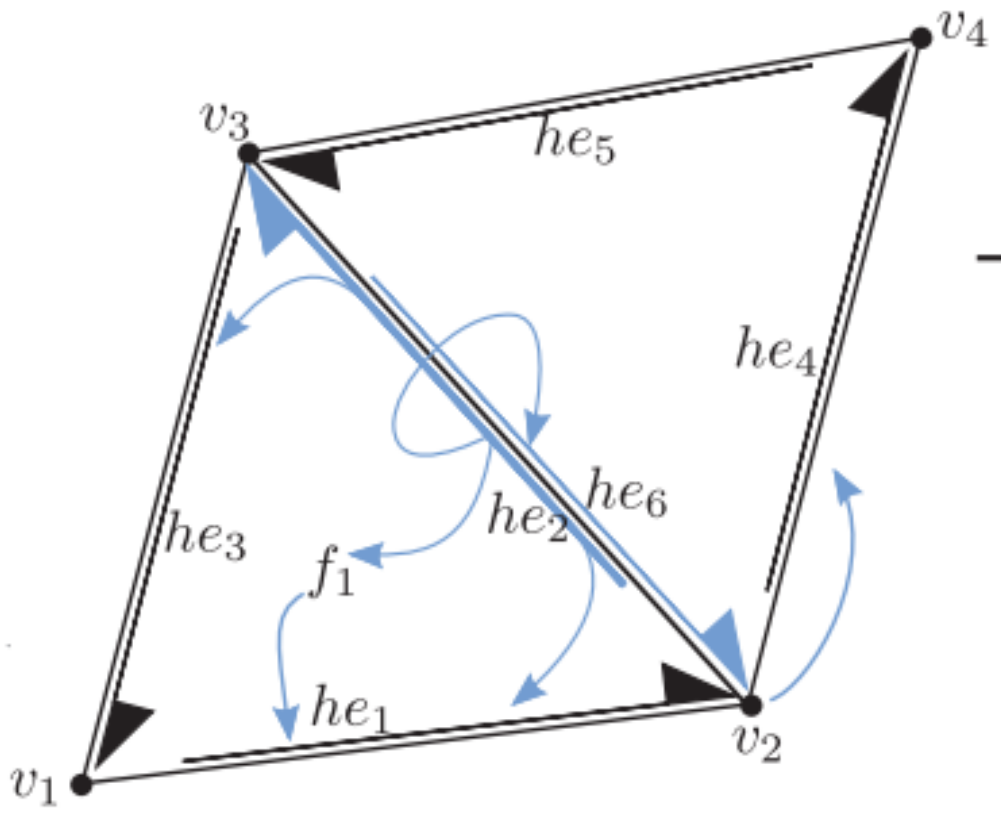
Face	*Edge*
0	a
1	c
2	d
3	a

Half-Edge

- Each half-edge stores:
  - A pointer to **previous** half-edge
  - A pointer to **next** half-edge
  - A pointer to other half-edge, on same edge (oriented in **opposite** direction)
  - Originating vertex
  - The face it is part of



Half-Edge



he_2		
$vert$	:	v_2
$prev$	:	he_1
$next$	:	he_3
opp	:	he_6
$face$	:	f_1

v_2		
$hedge$	:	he_4

f_1		
$hedge$	:	he_1

More Data Structures…

- Quad-edge
- Directed Edge
- Doubly Connected Edge List (DCEL)
- Corner tables

Mesh Data Structures Practice